

Simulation Setup: Subsonic Ablative Heat Wave in Gold

Two test cases are considered, differing only in the temporal power τ of the surface temperature drive. The medium is an ideal gas with polytropic equation of state $p = (\gamma - 1)\rho e$, where e is the specific internal energy. The Rosseland mean opacity governs the radiation diffusion. The simulation runs from $t = 0$ to $t_{\text{end}} = 2.0$ ns.

Case 1: $\tau = 0$

Material model. Adiabatic index $\gamma = \frac{5}{4}$, initial density $\rho_0 = 19.32$ g/cm³ (Gold).

$$e(T, \rho) = 3.4 \times 10^{13} \left(\frac{T}{\text{HeV}} \right)^{1.6} \left(\frac{\rho}{\text{g/cm}^3} \right)^{-0.14} \text{ erg/g}, \quad (1)$$

$$\kappa_R(T, \rho) = 7200 \left(\frac{T}{\text{HeV}} \right)^{-1.5} \left(\frac{\rho}{\text{g/cm}^3} \right)^{0.2} \text{ cm}^2/\text{g}. \quad (2)$$

Boundary conditions.

$$T_s(t) = 1 \text{ HeV}, \quad (3)$$

$$T_{\text{bath}}(t) = \left[1 + 0.169141 \left(\frac{t}{\text{ns}} \right)^{-\frac{79}{192}} \right]^{1/4} T_s(t). \quad (4)$$

Self-similar solution.

$$m_{\text{front}}(t) = 1.01889 \times 10^{-3} \left(\frac{t}{\text{ns}} \right)^{\frac{33}{64}} \text{ g/cm}^2. \quad (5)$$

Case 2: $\tau = \frac{158}{1285} \approx 0.12296$ (time-independent surface flux)

Same material model as Case 1 ($\gamma = \frac{5}{4}$, $\rho_0 = 19.32$ g/cm³, same e and κ_R), with a different drive.

Boundary conditions. The value of τ is chosen so that the self-similar radiation energy flux at the surface is time-independent ($c_S = 0$):

$$\tau = \frac{2 - 3\mu}{8 + 2\alpha + 2\beta + 3\beta\lambda - 3(4 + \alpha)\mu} = \frac{158}{1285}. \quad (6)$$

$$T_s(t) = \left(\frac{t}{\text{ns}} \right)^{\frac{158}{1285}} \text{ HeV}, \quad (7)$$

$$T_{\text{bath}}(t) = \left[1 + 0.204154 \left(\frac{t}{\text{ns}} \right)^{-\frac{632}{1285}} \right]^{1/4} T_s(t). \quad (8)$$

Self-similar solution.

$$m_{\text{front}}(t) = 8.34319 \times 10^{-4} \left(\frac{t}{\text{ns}} \right)^{\frac{193}{257}} \text{ g/cm}^2. \quad (9)$$